

Первое вычисление постоянной тонкой структуры на основе фундаментальных принципов

← Блог

БЛОГ

электромагнетизма, размер атомов и цвет окружающего мира. Голография с наблюдателем испускает его из одной ячейки экрана, которая должна согласовываться сама с собой.

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За большей частью видимой физики стоит небольшое число. Оно показывает, насколько сильно электрический заряд взаимодействует со светом. Оно задает масштаб атомной структуры, спектральных линий, химических процессов и большинства стабильных явлений, с которыми мы сталкиваемся каждый день. Физики называют его постоянной тонкой структуры.

Его низкоэнергетическое значение записывается как $\alpha^{-1}(0) \approx 137,036$. Люди измерили это число с точностью, которая посрамила бы большинство физических теорий. Чего не хватало, так это четкого пути от базовых принципов к самому числу.

Ричард Фейнман сформулировал эту проблему следующим образом:

«Чтобы вычислить постоянную тонкой структуры, равную $1/137$, нам понадобилась бы реалистичная модель практически всего сущего, а такой у нас нет».

Вокруг этого числа сложился целый фольклор. Говорят, Паули был одержим им. Эддингтон пытался его вывести, но потерпел неудачу. Лучше всего сохранить цитату Фейнмана, потому что в ней говорится о реальном препятствии. Чтобы вывести это число, нужна теория всего. Голографическая теория наблюдателя — именно такая теория, и с ее помощью это число можно вывести.

Одно предостережение следует сделать сразу. Чистый вывод по методу ортогональных проекций, без каких-либо измеренных входных данных в правой части, дает

$$\alpha^{-1} = 136.994\,835\,164\,621\,649\,457\,949\,994\,585\,787\,193\,262\,029\ldots$$

Измеренное значение константы, которая завершает построение низкоэнергетической квантовой хромодинамики, еще не представлено в виде символа.

$$\alpha^{-1}(0) = 137,035\,999\,177\,(21)$$

Разрыв резко локализован. Он находится в низкоэнергетическом адронном вкладе в конечную точку Томсона, которая, как известно, является сложным непertурбативным объектом в рамках квантовой хромодинамики. В выводе, основанном на методе оптической плотности, этот фрагмент не учитывается символически. Мы закрываем этот пробел с помощью измеренных данных $e^+e^- \rightarrow \text{адроны}$ в стандартной дисперсионной форме, которую используют все остальные.

Это чистый и структурированный результат. Архитектура исправлена. Упущение небольшое. Недостающая физика обозначена. В оставшейся части

поста я объясню, как работает вывод и почему он соответствует принципу «из первых рук».

Одна ячейка экрана, запрограммированная на самосогласование, фиксирует силу электромагнитного поля.

То, что читатель должен усвоить в первую очередь

Прежде чем приступить к алгебре, нужно разобраться с онтологией ОРН. Большинству физиков это не понравится. Но математика все равно требует этого.

Мы находимся внутри симуляции. Это не метафора. Это буквальная правда. Существует внешняя вселенная, динамика которой управляет голографическим экраном, и мир, который мы видим, включая пространство-время, частицы, поля и наблюдателей, читающих это предложение, — это то, что отображается на этом экране. ОРН — это теория о том, как работает отображение и каким требованиям оно должно соответствовать, чтобы вообще поддерживать существование наблюдателей.

The screen is built out of pixels. Each pixel is a finite cell with one dimensionless parameter that says how much area it occupies on the screen, in the natural units the outer dynamics provides. That parameter is the pixel area. We call it P .

P is the one dial of the simulation. Everything that lives on our side of the screen, every coupling, every mass ratio, every running scale, is emitted from P through the OPH source map. The fine-structure constant is one such emission. The strength of electromagnetism is what you get when you ask what value of P lets the simulation be self-consistent.

That is the move the rest of this post executes in detail. The pixel has to agree with itself when read in two ways. The value of P that satisfies the agreement fixes α .

The simulation does not get to choose 1/137. The consistency of the screen chooses it.

What the fine-structure constant actually is

The conventional formula is

$$\alpha = \frac{e^2}{4\pi\epsilon_0 \hbar c}$$

e is the elementary electric charge. ϵ_0 is the vacuum permittivity. $\hbar$ is the reduced Planck constant. c is the speed of light. Combine them in this exact way and the units cancel. Whatever you measure in, the number is the same.

That formula tells you what α measures. It says nothing about why the value comes out near 1/137. The "why" is the missing piece.

Most physicists prefer to quote the inverse, since 137.036 is easier to talk about than 0.007297.

The OPH starting point

OPH starts from one observation. No observer sees the whole universe. Each one has access to a finite local patch. Shared reality is assembled from the overlap relations between such patches.

At the smallest useful level, OPH models this with a single observer-supporting screen cell. The cell carries one key dimensionless parameter. Call it P . P is the realized local detuning of the cell.

If the cell sat at perfect self-similar balance, there would be no room for actual observation. No photons. No records. No measurements. The cell has to sit slightly off the balance point. The fine-structure derivation is the question of how far off.

EXPLAINER

Screen cell. The smallest local piece of the holographic screen that can support an observer. It carries one dimensionless detuning parameter and emits everything else through the OPH source map.

Why the golden ratio shows up

The first ingredient is the golden ratio

$$\varphi = \frac{1 + \sqrt{5}}{2}$$

In the OPH picture, the screen cell has a preferred self-similar balance point. That point is φ . This is not aesthetics. The entropy structure of the observer-supporting screen has a unique self-similar fixed point under the local recursion that builds the cell, and that fixed point is the golden ratio.

If the cell sat exactly at φ , the local screen would be in perfect equilibrium. Beautiful, and unobservable. A world that contains measurements requires a small disturbance away from that balance point. The disturbance is what photons live in.

Why $\sqrt{\pi}$ appears

The second ingredient is $\sqrt{\pi}$. It enters because the leading electromagnetic disturbance on the screen is shaped by a maximum-entropy boundary profile. That profile is a Gaussian, and the Gaussian normalization carries the familiar $\sqrt{\pi}$. The

width of the leading detuning around the equilibrium point is therefore $\sqrt{\pi}$ times the electromagnetic coupling.

The picture is concrete. There is a balance point at φ . There is a natural width scale for the disturbance. That is enough to write the first form of the fine-structure equation.

The same cell read in two ways

The whole derivation turns on one move. If you remember one structural step from this post, remember this one.

The same screen cell is read in two languages.

From the outside, the cell looks like a geometric displacement away from the golden-ratio equilibrium. The outer reading is

$$\text{outer reading : } P - \varphi$$

From the inside, the cell looks like an electromagnetic observation width: how wide a photon endpoint the cell can support. The inner reading is

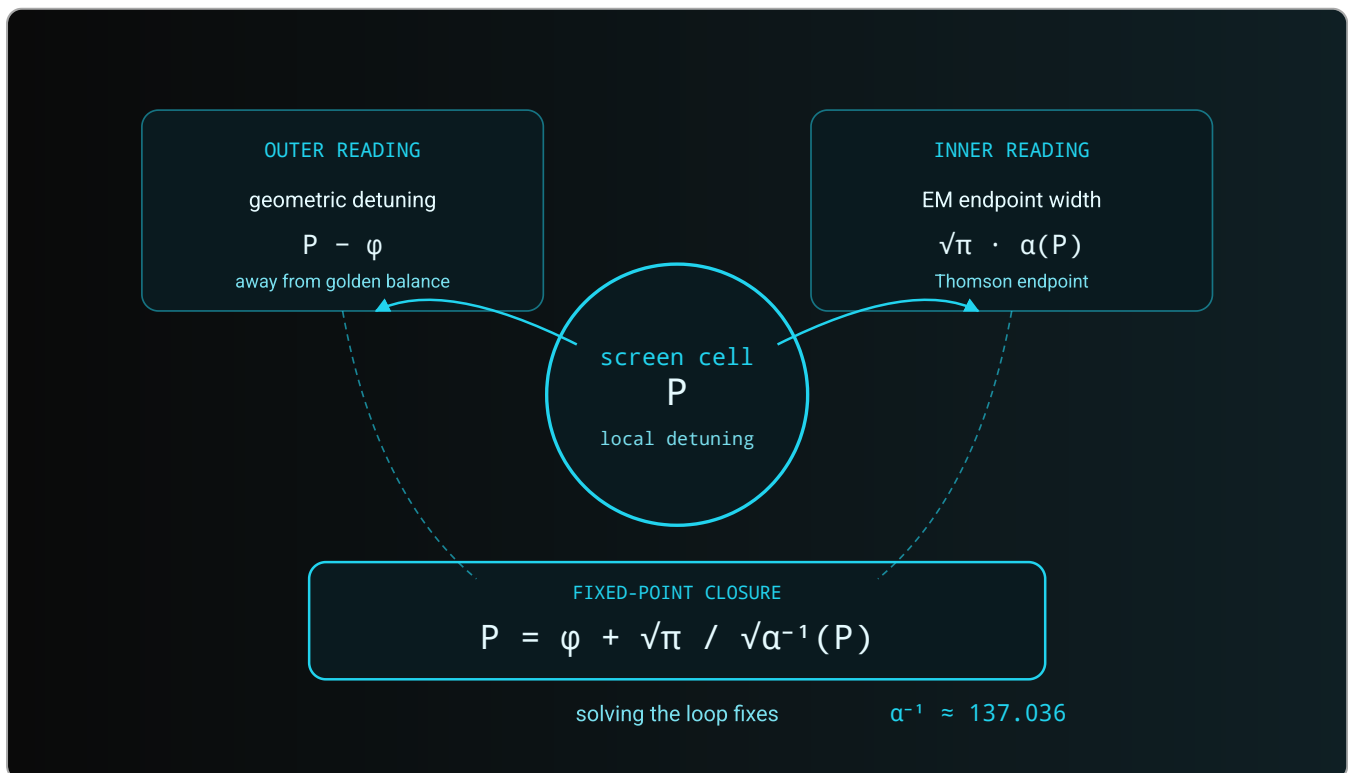
$$\text{inner reading : } \sqrt{\pi} \sqrt{\alpha(P)}$$

where the inside coupling is the inverse of the Thomson endpoint of electromagnetism on this cell.

The same object has to satisfy both readings at once. Setting them equal gives the OPH fixed-point equation:

$$P = \varphi + \frac{\sqrt{\pi}}{\sqrt{\alpha^{-1}(P)}}$$

This is the self-reference step. The cell is allowed to exist only if it agrees with itself when read geometrically and when read electromagnetically. There is one value of α that makes the loop close.



THE FIXED-POINT CLOSURE ON A SINGLE SCREEN CELL

Only one unique value agrees

The closure equation

$$P = \varphi + \frac{\sqrt{\pi}}{\sqrt{\alpha^{-1}(P)}}$$

is a consistency test. The left side is the pixel area we hypothesise. The right side is the pixel area the same pixel reports back about itself once you ask it what photons it can support. The pixel passes the test only if both sides equal each other.

Most pixel areas fail. Try one and watch.

Try $\alpha^{-1} = 120$. The right-hand side gives $\varphi + \sqrt{\pi/120} \approx 1.7798$. Now feed that pixel area back through the OPH source chain. The transport machinery returns $\alpha^{-1} \approx 156$, not 120. The pixel disagrees with itself. No world like this exists.

Try $\alpha^{-1} = 150$. The right-hand side gives $\varphi + \sqrt{\pi/150} \approx 1.7628$. Push that through the chain and the transport returns roughly $\alpha^{-1} \approx 128$. Disagreement again, in the other direction.

Try $\alpha^{-1} = 137$. The right-hand side gives $\varphi + \sqrt{\pi/137} \approx 1.7694$. The chain returns $\alpha^{-1} \approx 137.0$. The two readings nearly coincide. Sharpen the trial and the agreement sharpens with it.

Sweep α^{-1} across the entire real line and only one value satisfies the equation. That value, computed without any measured input, is

$$\alpha^{-1} = 136.994\,835\,164\,621\,649 \dots$$

Every other choice produces a pixel whose two faces report different areas. A universe built on such a pixel would not have stable photons, would not have a reproducible Coulomb law, would not have atoms. The world we measure exists at exactly the unique value of electromagnetic strength that lets the pixel be what it is.

How the actual computation finds it

The equation is too tangled for a closed-form solution. The transport function t_Q involves the full source chain, two-loop renormalisation-group running for the

gauge couplings, and the leptonic and electroweak transport kernels. Solving by hand is out of the question. The OPH stack uses a numerical solver, and a very simple one is enough, because the closure map is a contraction.

A contraction map is a function that pulls every input toward a single fixed point. Plug any reasonable starting guess into the right-hand side. The output is closer to the answer than the input was. Plug the output back in. Closer still. The error roughly halves every iteration in the relevant range. After a handful of rounds, the value stops moving in the digits you care about. After a few dozen rounds, it stops moving in the fortieth decimal.

That is what produces the long string. Banach's fixed-point theorem guarantees the iteration converges to the unique value where input equals output. The OPH solver is a direct implementation of that iteration with arbitrary-precision arithmetic on the source chain. It is not a search through a parameter space. It is a settling process driven by the consistency condition itself.

The next section walks through the iteration step by step.

The source map

The fixed-point equation is only useful once the Thomson endpoint can be computed from a trial value of P . That requires leaving the geometric cell picture and entering particle physics proper.

The OPH source chain begins with the source map

$$M_U(P) = E_P \exp\left(-2\pi P^{1/6}\right)$$
$$E_{\text{cell}}(P) = \frac{E_P}{\sqrt{P}}$$

E_P is the Planck energy. The first quantity is the unification-scale readout of the cell. The second is its characteristic energy. From one local number, the cell starts emitting an entire physics package.

Gauge couplings and the electroweak anchor

The next relation is the pixel closure equation, which fixes the unified coupling and the running gauge family from the same local cell. The electroweak and strong sectors are not adjusted independently. They are emitted from one shared local scale.

From the running family, OPH forms the electroweak anchor at the Z mass:

$$a_0(P) = \alpha_2^{-1}(m_Z; P) + \frac{5}{3} \alpha_1^{-1}(m_Z; P)$$

This is the inverse electromagnetic coupling at the electroweak anchor scale. It is the source-side starting point for the low-energy endpoint. The famous textbook 1/137 is not this number. It lives at zero momentum, far below the Z. To get there, one transports the coupling down.

Transport down to the Thomson endpoint

OPH then uses a Ward-projected transport theorem on the unbroken electromagnetic branch:

$$\alpha_{\text{em}}^{-1}(q^2; P) = \frac{8\pi^2}{t_Q(q^2; P)}$$

The low-energy fine-structure constant is the zero-momentum endpoint of this transport family:

$$\alpha_{\text{Th}}^{-1}(P) = \lim_{q^2 \rightarrow 0} \alpha_{\text{em}}^{-1}(q^2; P) = a_0(P) \frac{t_Q(m_Z^2; P)}{t_Q(0; P)}$$

That endpoint is the quantity that feeds the closure. The full equation is

$$P = \varphi + \frac{\sqrt{\pi}}{\sqrt{\alpha_{\text{Th}}^{-1}(P)}}$$

This is the OPH route from one local cell to the value of the fine-structure constant.

Computing the number, step by step

All of the above is a closed system of equations in one unknown, the pixel area P . The fine-structure constant comes out by solving the loop. Here is the actual procedure that lands at **136.994 835 164 621 649 . . .**

Step 1. Pick a trial pixel area. Start with any sensible P , for example $P_0 = \varphi$. The loop converges from a wide basin, so the seed barely matters.

Step 2. Run the source map. Compute the unification scale and the cell scale from the trial:

$$M_U(P) = E_P e^{-2\pi P^{1/6}}, \quad E_{\text{cell}}(P) = \frac{E_P}{\sqrt{P}}$$

Step 3. Close the gauge couplings. Solve the OPH pixel-closure equation

$$\bar{\ell}_{SU(2)}(t_2(P)) + \bar{\ell}_{SU(3)}(t_3(P)) = \frac{P}{4}$$

for the unified coupling $\alpha_U(P)$. Run the standard two-loop renormalisation-group equations down from $M_U(P)$ to the Z mass to get the gauge family $\alpha_1(m_Z; P), \alpha_2(m_Z; P), \alpha_3(m_Z; P)$.

Step 4. Form the electroweak anchor.

$$a_0(P) = \alpha_2^{-1}(m_Z; P) + \frac{5}{3} \alpha_1^{-1}(m_Z; P)$$

This is the inverse electromagnetic coupling at $q^2 = m_Z^2$.

Step 5. Transport down to the Thomson endpoint. Evaluate the Ward-projected transport time at the anchor and at zero momentum, then take the ratio:

$$\alpha_{\text{Th}}^{-1}(P) = a_0(P) \frac{t_Q(m_Z^2; P)}{t_Q(0; P)}$$

In the pure derivation, the transport time is built from the leptonic transport contribution, the electroweak remainder, and the electroweak anchor. The low-energy hadronic spectral term that the OPH chain does not yet emit symbolically is

set to zero, and that is the only place the pure derivation and the measured value differ.

Step 6. Plug into the closure.

$$F(P) \equiv \varphi + \frac{\sqrt{\pi}}{\sqrt{\alpha_{\text{Th}}^{-1}(P)}}$$

Step 7. Iterate to the fixed point. Set $P_{n+1} = F(P_n)$ and repeat steps 2 through 6. The map is a contraction in this region, so successive iterates settle very fast onto a fixed point $P_\star$ with $F(P_\star) = P_\star$. The pure derivation lands at

$$P_\star = 1.630\,972\,095\,694\,329\,018\ldots$$

Step 8. Read off the constant. Substitute $P_\star$ back into the closure rearranged for the coupling:

$$\alpha^{-1} = \frac{\pi}{(P_\star - \varphi)^2}$$

That single line is the equation that outputs the number. Plug in the fixed point and you get

$$\alpha^{-1} = 136.994\,835\,164\,621\,649\,457\,949\,994\,585\,787\,193\,262\,029$$

When we plug in the measured hadronic spectral term inside $t_Q(0; P)$, supplied by $e^+e^- \rightarrow \text{hadrons}$ data through the standard dispersion integral, the fixed point shifts to $P_\star = 1.630\,968\,209\dots$ and the same readout equation gives $\alpha^{-1}(0) = 137.035\,999\,177\,(21)$.

That is the whole computation. One unknown. One contraction. One readout equation that turns the pixel area into the strength of electromagnetism.

Two numbers, two meanings

Two values matter, and they should never be confused.

The pure derivation, computed entirely from the OPH equations with no measured input on the right-hand side, gives

$$\alpha^{-1} = 136.994\,835\,164\,621\,649\dots$$

The same equations, with the one missing low-energy hadronic term supplied by measured data, give

$$\alpha^{-1}(0) = 137.035\,999\,177\,(21)$$

The difference between the two is about 0.041 in inverse- α units. Small, and concentrated in one specific place.

Where the missing piece lives

The Thomson endpoint splits into four contributions:

$$A_T(P) = a_0(P) + \Delta_{\text{lep}}(P) + \Delta_{\text{had}}(P) + \Delta_{\text{EW}}(P)$$

The source-side anchor, the leptonic transport term, and the electroweak remainder are all emitted by the OPH chain. The hadronic spectral term is the hard one. It is the contribution of confined quarks and gluons to the electromagnetic vacuum polarization at low energy. It depends on the hadronic spectral measure, which is exactly the part of QCD that resists clean symbolic treatment.

That term contains resonances, thresholds, strong correlations, finite-volume effects, continuum limits, and normalization issues. It is the same place every other theory hits a wall. The OPH derivation does not pretend the endpoint is closed in pure form. The missing object is named directly: the Ward-projected hadronic spectral measure. Producing it symbolically is a heavy nonperturbative QCD problem and is work in progress.

The slight mismatch is therefore not embarrassing. It is exactly where any serious derivation of α would hit the roughest part of the Standard Model.

Why using measured hadronic data is legitimate

The closure rule is simple. Keep the pure OPH derivation visible as one number. Use measured $e^+e^- \rightarrow \text{hadrons}$ data only for the one missing low-energy hadronic term, and report the resulting value separately.

The standard dispersion relation for the hadronic vacuum-polarization contribution is

$$\Delta\alpha_{\text{had}}(q^2) = -\frac{\alpha q^2}{3\pi} \text{P.V.} \int ds \frac{R(s)}{s(s - q^2)}$$

with $R(s)$ the measured ratio of the bare $e^+e^- \rightarrow \text{hadrons}$ cross section to the pointlike $e^+e^- \rightarrow \mu^+\mu^-$ one. This is exactly what every precision program plugs in.

Plugging measured data into one isolated spectral integral is a different operation from typing in the measured final value of α . The fixed-point architecture, the source frame, and the nonhadronic pieces are all derived. The empirical input is one specific term on a specific line, in the same form everyone else in precision QED uses.

A true first-principles derivation

The derivation does not begin with the measured value of α . It begins with a local observer-supporting screen cell, a self-similar equilibrium point, a maximum-entropy boundary normalization, the OPH source map, the gauge-coupling closure, the Ward-projected electromagnetic transport theorem, and the fixed-point self-consistency relation.

Out of that, one number falls. To 0.03 percent of the measured value, with a single named missing term in low-energy QCD that closes the rest.

Compare that to the prior state of the art. The fine-structure constant is the single best-measured number in physics, and the standard answer to "where does it come from" is "we measure it." Eddington's numerology lasted a few years.

Feynman declared the problem unsolvable without a theory of everything. OPH is a theory that emits the standard model from observer-patch consistency, and it carries the derivation through to the endpoint.

The picture in one paragraph

The local observer-supporting cell wants to sit at the golden-ratio balance point. Observation requires a small electromagnetic disturbance away from that exact equilibrium. The same cell is read from the outside as a geometric detuning and from the inside as an electromagnetic endpoint width. Forcing those two readings to agree gives the fixed-point equation. The pure OPH derivation lands at $\alpha^{-1} = 136.9948 \dots$. The remaining gap to the measured **137.036** is one specific low-energy hadronic term that requires a nonperturbative QCD backend. Plugging in measured data closes that one piece, in the same dispersion form precision QED already relies on.

The structure is in place. The missing physics is known. The pure derivation lands right next to the measured value. That is what a derivation of α from first principles looks like.

References

- [1] R. P. Feynman, "QED: The Strange Theory of Light and Matter", Princeton University Press, 1985.
- [2] CODATA recommended value of the fine-structure constant, 2022 adjustment.
- [3] B. Mueller, K. Xue, K. A. Anirudha, "Reality as a Consensus Protocol".
- [4] B. Mueller, K. Xue, "Screen Microphysics and Observer Synchronization in OPH".
- [5] OPH particle paper and supporting notes, paper stack.
- [6] F. Jegerlehner, "The hadronic contribution to the anomalous magnetic moment of the muon and to the running fine structure constant".

Go Deeper

Follow the derivation, explore the source stack, or read the rest of the architecture.



Start Here

The guided learning track. The cleanest way into OPH if this article sparked your curiosity.



OPH Lab

Interactive derivations, including the source-side particle physics chain.



The Standard Model Chapter

How OPH emits the gauge structure, the running couplings, and the electroweak anchor.



The Paper Stack

The particle paper and the rest of the technical record.